Practical Manual
Fermentation Technology BT612P



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Practical 1: Media formulation (yeast)

Introduction

Culture media, also known as growth media, are specific mixtures of nutrients and other substances that support the growth of microorganisms such as bacteria and fungi (yeasts and molds). Culture media are used for quality control tests of nonsterile raw materials and finished products as well as for microbial contamination (sterility) tests in applications such as hygiene monitoring, sterilization process validation and determination of the effectiveness of preservatives and antimicrobial agents. All micro-organisms require water, sources of energy, carbon, nitrogen, mineral element and vitamin plus oxygen in their growth medium. On a small scale, it is simple to device a medium containing pure compounds, but the resulting medium although satisfy the growth, may be unsuitable for use in a large scale process.

The essential part of media designing is to be considering the stoichiometry of growth and formation of desired product which is given below:

Carbon and energy source + Nitrogen source + O_2 + Other requirements ----> Biomass + Products + CO_2 + H_2O+ Heat Essential characteristics of formulated media

1. It should produce the maximum yield, maximal concentration with maximal rate of production of desired biomass, of consistent quality which is always readily available.
2. It should also have minimal problems during media designing.
3. It should have minimal problems during recovery of desired product/ biomass especially during aeration and agitation, extraction, purification and waste treatment.

Basic media constituents

1. Water

It is major component of all fermentation media. Following factors need to be considered about water for media preparation

- pH
- Dissolve salt
- Effluent contamination

Mineral water content of water is very important in brewing, and most critical in the mashing process, and historically influenced the siting of breweries and the type of beer produced.

2. Energy Sources

Energy for growth comes from either the oxidation of medium components or from light. Most industrial micro-organisms are chemo-organotrophs, therefore the commonest sources of energy will be the carbon source such as carbohydrates, lipids and proteins.

3. Carbon sources

Carbon requirement for the medium is normally provided by:

- **Sucrose**: Sugarcane, sugar beet molasses
- **Glucose**: Corn sugar, starch, cellulose
- **Lactose**: Milk whey
- **Fats**: Vegetable oil
- **Starch**: Maize grains, cereals, potatoes and cassava
- **Hydrocarbons**: Petroleum fractions

4. Nitrogen Sources

Microorganisms generally can use inorganic or organic N.

- Inorganic sources: ammonia, ammonium salts
- Organic sources: amino acid, proteins and urea, Corn steep liquor, Yeast extract, Peptones, Soya bean meal.

5. Oxygen sources

Oxygen is always provided in water.

Some organisms require molecular oxygen as terminal oxidizing agents to fulfill their energetic needs through aerobic respiration. These organisms are obligatorily aerobic.

For obligate anaerobes molecular O_2, is a toxic substance.

Some organisms are facultative anaerobes and can grow with or without molecular 02.

Culture media can be distinguished on the basis of composition, consistency and purpose.

Classification of culture media on the basis of consistency

1. **Solid medium**

   solid medium contains agar at a concentration of 1.5-2.0% or some other, mostly inert solidifying agent. Solid medium has physical structure and allows bacteria to grow in physically informative or useful ways (e.g. as colonies or in streaks). Solid medium is useful for isolating bacteria or for determining the colony characteristics of the isolate.

2. **Semisolid media**

   They are prepared with agar at concentrations of 0.5% or less. They have soft custard like consistency and are useful for the cultivation of microaerophilic bacteria or for determination of bacterial motility.

3. **Liquid (Broth) medium**

   These media contain specific amounts of nutrients but don't have trace of gelling agents such as gelatin or agar. Broth medium serves various purposes such as propagation of large number of organisms, fermentation studies, and various other tests. e.g. sugar fermentation tests, MR-VR broth.

Classification of culture media based on the basis of composition

1. Synthetic or chemically defined medium

A chemically defined medium is one prepared from purified ingredients and therefore whose exact composition is known.

2. Non synthetic or chemically undefined medium

Non-synthetic medium contains at least one component that is neither purified nor completely characterized nor even completely consistent from batch to batch. Often these are partially digested proteins from various organism sources. Nutrient broth, for example, is derived from cultures of yeasts.

Medium formulation (Yeast)

YEPD or yeast extract peptone dextrose, also often abbreviated as YPD, is a complete medium for yeast growth. It contains yeast extract, peptone and glucose or dextrose. It can be used as solid medium by including agar. The yeast extract will typically contain all the amino acids necessary for growth. YEPD is used as a growth medium to grow yeast cultures.

Composition (g/L)

Peptone 20

Yeast extract 10

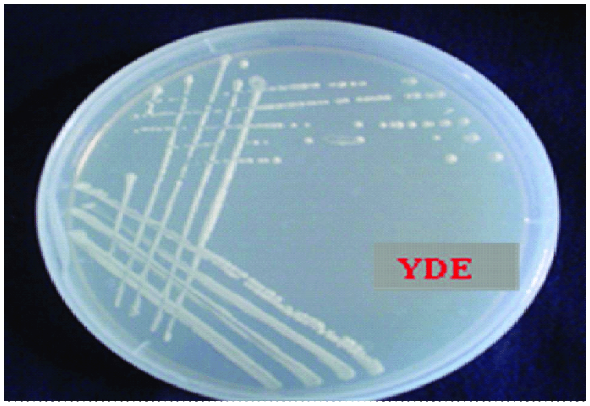
Dextrose 20

pH:4-4.5

1. Add peptone 2.0 grams, Yeast extract 1.0 g and dextrose 2.0 g in 50 ml distilled water. Adjust the pH to 5.5 with 0.1 N phosphoric acid.

2. Make up the volume to 100 ml with distilled water.

3. Transfer medium to 250 ml Erlenmeyer flask and cotton plug it. Sterilize by autoclaving at 15 psi, 121°C for 15 minutes.



Potato dextrose agar and potato dextrose broth are common microbiological growth media made from potato infusion, and dextrose. Potato dextrose agar (abbreviated "PDA") is the most widely used medium for growing fungi.

Potato infusion can be made by boiling 200 grams of sliced (washed but unpeeled) potatoes in ~ 1 litre of distilled water for 30 minutes and then decanting or straining the broth through cheesecloth. Distilled water is added such that the total volume of the suspension is 1 litre. PDA is also available commercially as premixed medium.

Composition (g/L) of PDA

- Potato infusion 200 g
- Dextrose 20 g
- Agar 15 g
- pH:5.5

Malt Extract Agar is used as a general purpose growth media to isolate and cultivate molds from clinical samples, as well as a wide range of environmental sources. Malt Extract Agar is designed to contain the proper formulation of carbon, protein and nutrient sources essential for yeast and mold growth. Dextrose is added to the medium to provide a carbon and energy source for fungi. Additionally, Malt Extract Agar contains digests of animal tissues (peptones) which provide a nutritious source of amino acids and nitrogenous compounds for the growth of mold.

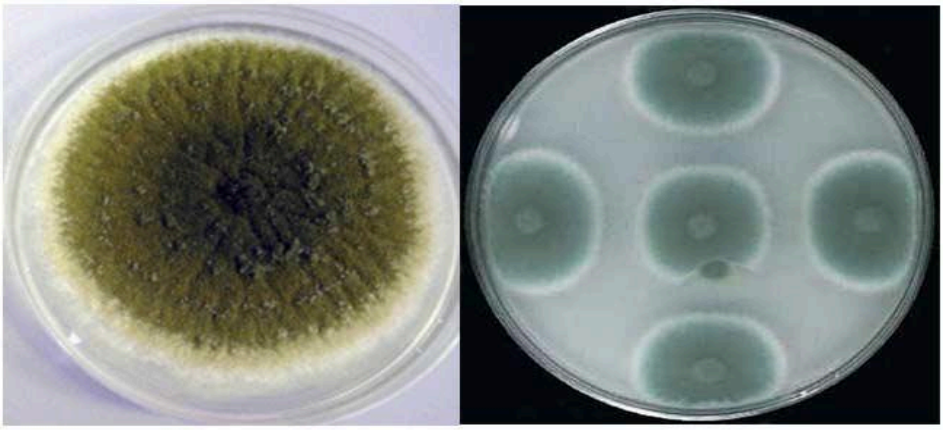
The pH is adjusted to approximately 5.5 in order to enhance the growth of fungi and to slightly inhibit bacterial growth commonly found as environmental contaminants.

Composition of Malt Extract Agar Medium

- Malt Extract 20.0gm
- Dextrose 20.0gm
- Peptone 6.0gm
- Agar 15.0gm

Procedure (Preparation of PDA)

1. Add 3.9 grams in 50 ml distilled water. Adjust the pH to 5.5 with 0.1 N phosphoric acid.
2. Heat to boiling to dissolve the medium completely. Make up the volume to 100 ml with distilled water.
3. Transfer medium to 250 ml Erlenmeyer flask and cotton plug it.
4. Sterilize by autoclaving at 15 psi, 121°C for 15 minutes.



***Aspergillus flavus* on PDA agar** ***Penicillium chrysogenum* on PDA**

Practical 2:

Development of Conidial Inoculum for Fungal Fermentation

Principle

Before a culture is used in a production fermenter it goes through a process of preparation and propagation which is referred to in this study as inoculum development. Al though it has always been recognized that inoculum type and quality have a considerable effect on the course of a fermentation. The inoculum is typically prepared in a stepwise sequence employing increasing volume of media.

Factors of importance

A number of factors need to be considered to ensure that inoculum development is a satisfactory process.

Type of Inoculum

With bacteria, rapidly growing cultures are often preferred. Cultures preserved as spores are often subjected to a heat treatment known as "heat shock" to induce spore germination and obtain vegetative growth. With filamentous microorganisms such as fungi and actinomycetes, inocula may grow as pellets or as free mycelium. The type of inoculum dictates how inoculum build-up should be initiated.

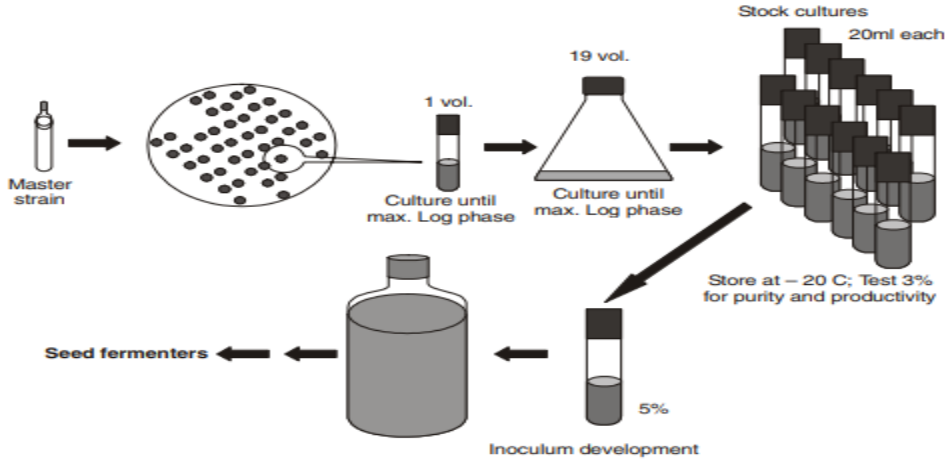
Size of Inoculum

All steps of the inoculum development process, except that of the initial inoculation from a stock culture, require the transfer of approximately 0.5 to 0.1 percent inoculum by volume from the preceding step in the sequence. At each step, the organisms should grow quickly and in high numbers so that the period of incubation required is relatively short. Higher levels of inocula are commonly used in the production scale and may require separation of the inoculum from the medium. It should be insured that high levels of inoculum do not concurrently cause excessive dilution of the production medium, unwanted pH changes or carryover of unwanted nutrients or metabolic waste products.

Stability of Inoculum

No physiological or genetic change should occur during the sequence of transfers of microorganisms. However, although mutation does sometimes occur, it is usually infrequent and lethal to the cell, or at least provide no selective advantage over the non-mutated cells. However,

problems may occur if the organism used for the fermentation is in itself a mutant strain. Mutant cells are not always stable and the frequency of back mutation might be high.



Inoculum for fungal fermentations

Majority of industrially important fungi are capable for asexual sporulation, so it is common practice to use a spore suspension as seed during an inoculum development programme. In some fermentations, large-scale inoculum must consist of spores. To obtain a spore crop, the preserved culture is cultivated on a solid substrate in 2-10 liter glass vessels under conditions of constant temperature and sterile aeration for 8-24 days. Substrate for production of large amounts of spores is a granular material such as bran, peat, rice, or barley. In order to ensure continued aeration, the substrate must be shaken daily, which makes maintenance of aseptic conditions difficult. When considering production of gluconic acid by *Aspergillus niger,* the merits of inoculating the final fermentation directly with a spore suspension as compared with germinating the spores in a seed tank to give a vegetative inoculum. Direct spore inoculation would avoid cost of installation and operation of the seed tanks. However, use of germinated spores reduced fermentation time of final stage, thus allowing a greater number of fermentations to be carried out per year.

Procedure

1. Some fungi will not produce asexual spores and therefore an inoculum of vegetative mycelium must be used. *Aspergillus niger* is a fungus (mold) used for commercial production of various enzymes.
2. Cultures grow on long slants (25 X 150 mm test tubes) of potato dextrose agar for 1 week at 24°. Growth from three slants scrape off and transfer to a 9-dm3 carboy containing 4 dm3 of a liquid medium composed of 2% glucose, 0.3% MgSO4.7H20, 0.3% NH4CI and 0.3% KH2PO4.
3. The medium aerated for 75 hours at 28°before transfer to a 100-dm^3 seed fermenter containing the same medium. Major problem in using vegetative mycelium as initial seed is difficulty obtaining a uniform, standard inoculum.
4. The procedure may be improved by fragmenting the mycelium in a homogenizer, such as a Waring blender, prior to use as inoculum. This method provides a large number of mycelial particles and therefore a large number of growing points.

Practical 3: Solid-state fermentation in 250ml flask using wheat bran as substrate

Solid-State Fermentation (SSF)

In solid-state fermentation, the microorganisms grow on a moist solid with little or no 'free' water, although capillary water may be present. Examples of this type of fermentation are seen in mushroom cultivation, bread-making and the processing of cocca, and in the manufacture of some traditional foods, e.g. miso (soy paste), soy sauce, tempeh (soybean cake) and gari (cassava), which are now produced in large industrial operations. SSF is also called as Koji Fermentation.

Advantages of solid state fermentation

1. Cheap
2. environmental friendly
3. High yield
4. Low availability of water reduces possibilities of contamination

Disadvantages

1. Downstream processing difficult
2. difficult to scale up
3. difficult to monitor and control different parameters

Materials Required

Beaker, Stirrer, 250 ml Erlenmeyer flask, cylinder, wheat bran, Zn $SO_4.7H_2O$, $FeSO_4.7H_2O$, Cu $SO_4.7H_2O$

Procedure

1. Add ten grams of solid substrate such as wheat bran in 250 ml Erlenmeyer flask.
2. Moisten the substrate with 10 ml of suitable diluent (containing mg/L: Zn $SO_4.7H_2O$, 6.2; $FeSO_4.7H_2O$, 6.8; Cu $SO_4.7H_2O$, 0.8; Distilled water, 1000 ml).
3. Cotton plug the flask and Sterilize the flask in an autoclave at 121°C, 15 psi for 15 minutes.
4. Cool the flask at room temperature.

5. Add one ml of the bacterial suspension to flask.

6. Incubate the flask at 30/37°C for 48-72 h.

7. After fixed period of incubation, add 100 ml of distilled water to each flask containing fermented bran.

8. Place the flask in incubator shaker at 160 rpm for one hour.

9. After one hour, filter the contents of the flasks and use the filtrate for the estimation of enzyme (alpha amylase).

Practical 4:

Shake flask fermentation in 250ml flask using suitable medium

Shake flask fermentation

The proper procedure for a fermentation is first to inoculate a small flask of nutrient broth with a pure culture from a Petri dish, a culture tube (containing liquid nutrient), or a slant tube (containing solid gel). The inoculated flask is constantly agitated in a temperature controlled flask shaker. A small amount of the culture in the original flask is pipetted out during the *exponential growth phase*, or *log phase*, and is used to inoculate the next flask. This process is repeated a few times to ensure that the culture is acclimated before it is employed to study the fermentation kinetics. A similar process of repeated inoculation is carried out in the fermentation industry to build up enough inoculum needed to seed a larger fermenter. It is used for pickling vegetables, brewing beer and producing wine and soy sauce and production of enzymes.

Materials Required

Beaker, Stirrer, 250 ml Erlenmeyer flask, cylinder, agar

Procedure

1. Add molasses medium I in 250-mL Erlenmeyer flasks. Molasses medium I contain 900 mL molasses with 180 g L^{-1} sugar content and 100 mL malt and 2 g L^{-1} magnesium sulphate. Molasses medium II contain molasses with final sugar content of 200 g L^{-1}, 2.3 g L^{-1} diammonium phosphate, 0.8 g L^{-1} ammonium sulphate and 0.5 mL antifoam,
2. Sterilize all media by cotton plug the flask and sterilize the flask in an autoclave at 121°C, 15 psi for 15 minutes.
3. Cool the flask at room temperature and inoculate with the desired volume of inoculum.
4. For inoculum preparation the reference *Aspergillus niger* strain use.
5. Prepare serial dilutions and culture on MDA and PDA agar plates. The plates incubate at 30 °C for two to three days
6. Adjust the initial pH to 4.5 and incubate the flasks at 30 °C with shaking at 200 rpm.

7. Monitor growth by measuring the absorbance at 600 nm through time intervals of 2 h using a spectrophotometer
8. Record absorbance and convert A_{600} records to biomass, express as g L^{-1} fresh weight, based on a standard curve.
9. Limit the Aeration to allow maximum oxygen exchange efficiency, agitation and mixing of the culture components
10. Allow Growth for 16 h and fresh weights determine
11. After an incubation period, separate the cells by centrifugation and determine the residual sugar concentration in the liquids using a saccharometer

Practical no: 05

Static culture/fermentation in 250ml flask using suitable medium for Aspergillus niger.

Ingredients and Materials:

- Medium: Potato Dextrose Agar (PDA) or Czapek-Dox Agar are commonly used for Aspergillus niger. For liquid culture, you can prepare a similar medium in liquid form.
- 250 ml Erlenmeyer flask with cotton plug or foam stopper
- Inoculating loop or needle
- Aspergillus niger spores (obtained from a pure culture)

Preparation of Medium:

1. **Potato Dextrose Broth (PDB) Medium**:

    - Prepare PDB medium by dissolving 24 g of potato dextrose broth powder in 1 liter of distilled water. Adjust the pH to around 5.6-6.0 using HCl or NaOH if necessary.
    - Autoclave the medium at 121°C for 15-20 minutes to sterilize.

2. **Czapek-Dox Broth Medium**:

    - Prepare Czapek-Dox broth by dissolving the components (sodium nitrate, potassium chloride, magnesium sulfate, dipotassium phosphate, and sucrose) in distilled water according to the recipe. Adjust pH to 6.0-6.5.
    - Autoclave the medium at 121°C for 15-20 minutes.

Inoculation and Incubation:

1. **Inoculation**:

    - After sterilization and cooling of the medium to about 45-50°C, aseptically transfer a small amount of Aspergillus niger spores using an inoculating loop or needle into the flask containing the prepared medium.
    - For liquid cultures, inoculate the spores directly into the sterile broth.

2. **Incubation**:

    - Seal the flask with a cotton plug or foam stopper to allow for gas exchange while preventing contamination.
    - Incubate the flask statically at the appropriate temperature for Aspergillus niger growth, typically around 25-30°C.
    - Monitor the growth of the fungus over time, which may vary depending on the specific strain and conditions.

Notes:

- **Temperature**: Aspergillus niger grows well at temperatures between 25-30°C.
- **Sterility**: Ensure all equipment and media are sterilized properly to avoid contamination.
- **Observation**: Check for growth characteristics such as color, texture, and odor to assess the culture's progress.

Practical no: 06

Medium for Aspergillus niger (Fungi)

1. Potato Dextrose Agar (PDA) Medium:

Ingredients (per liter of distilled water):

- 24 g Potato Dextrose Agar (PDA) powder

Preparation:

1. Dissolve 24 g of PDA powder in 1 liter of distilled water.
2. Mix well and heat to dissolve completely.
3. Adjust the pH to around 5.6-6.0 using HCl or NaOH if necessary.
4. Autoclave at 121°C for 15-20 minutes to sterilize.
5. Pour into Petri dishes or use in flasks/bottles for agar slants.

Use:

- PDA is suitable for both agar plates and slants. It provides a rich source of nutrients and supports good growth of Aspergillus niger, allowing for colony morphology observation and spore production.

2. Czapek-Dox Agar Medium:

Ingredients (per liter of distilled water):

- 30 g Sucrose
- 3 g Sodium nitrate
- 1 g Dipotassium phosphate
- 0.5 g Magnesium sulfate
- 0.5 g Potassium chloride
    - g Ferrous sulfate
- 15 g Agar (for solid medium)

Preparation:

1. Dissolve the sucrose, sodium nitrate, dipotassium phosphate, magnesium sulfate, and potassium chloride in 1 liter of distilled water.
2. Adjust the pH to 6.0-6.5 using HCl or NaOH if necessary.
3. Add agar (if making solid medium) and heat to dissolve completely.
4. Autoclave at 121°C for 15-20 minutes to sterilize.
5. Pour into Petri dishes or use in flasks/bottles for agar slants.

Use:

- Czapek-Dox Agar is suitable for studying fungal growth and metabolic activities. It provides a more defined medium compared to PDA and is useful for specific research purposes where precise nutrient composition is desired.

Notes:

- **Sterility**: Ensure all equipment and media are sterilized properly to avoid contamination.
- **Temperature**: Aspergillus niger typically grows well at temperatures between 25-30°C.
- **Incubation**: Incubate plates or flasks in a suitable environment (e.g., in a microbiological incubator) and monitor growth periodically.

Practical 7:

Fermentation of sugars

Anaerobic fermentation occurs in the fermentation vessel once the oxygen is discharged and replaced with N_2, CO_2, or another by-product of the fermentation process. Anaerobic fermentation is usually a slower process. In the mid-1850s, the French chemist Louis Pasteur produced anaerobiosis by boiling the medium to drive out oxygen and then introducing inert gas for cultivation.

Theory:

Yeasts are tiny, microscopic organisms—or microorganisms—that are actually a type of fungus. This means that they are more closely related to a mushroom than to plants and animals or bacteria (the latter of which are also microorganisms). These little critters might sound strange and different, but people have been using them for thousands of years to make bread rise. They turn this food into energy and release carbon dioxide gas as a result. This process is known as fermentation. When yeasts eat sugar and turn it into energy, they also produce carbon dioxide. This process is known as fermentation. In this activity, the balloons on the bottles should have captured carbon dioxide produced by the yeasts during fermentation.

Apparatus Required:

Beaker 500ml, 4 flasks of 250 ml, burner, tripod stand.

Reagents Required:

Yeast powder, water, Sugar.

Procedure:

- Take 5 flask and mark them A, B, C, D and E.
- Take 2.5g of yeast and put it in flask B, C and D.
- After that add 5g of sucrose in flask A, C and E.
- Add 10 gram of sucrose in flask D.
- Now add 100ml of distilled water to all flasks.
- Now boil only flask C for 5 minutes.
- Now close the mouth of all flasks with balloons.
- Leave the whole assembly for 24 hours.

Results:

Balloons of flask A, B and C doesn't blown but the balloons of flask D and E will blow. This shows that fermentation occur in flask D and E during which CO2 evolved and become reason of blowing of balloons.

Practical 8: Growth Kinetics

Introduction:

Bacteria normally reproduce by binary fission, forming two equal-sized progeny cells and thus doubling their number with each division. This type of cell division is called exponential; i.e., one cell divides to form two, each of these cells divides so that four cells are obtained, and so forth in a geometric progression. The unit of microbial growth is "generation time", which is the time required to achieve a doubling of the population size and is designated as tgen. Generation

time can be estimated by determining cell numbers during the period of active cell division and is expressed mathematically as:

$$tgen = t \ln 2 / \ln Xt - \ln X0 = t \log 2 / \log Xt - \log X0$$

In the above equations, t is the elapsed time during which growth is measured, and Xo and Xt are the number of bacteria at times zero and t, respectively. The presence of 2 in the numerator of the equations is due to the fact that when doubling of the population number occurs, the quantity Xt/Xo (called the doubling constant) will be equal to 2. The generation time of many bacteria is usually several hours. However, under optimum conditions, E. coli has a generation time of about 20 minutes. Thus it can be calculated that a single E. coli cell will produce about a thousand progeny in 3.3 hours and over a million in 6.6 hours. If the generation time and the elapsed time are known, then the number of generations can be estimated as follows:

$$\text{Number of generations} = t / tgen$$

A bacterial population follows a characteristic growth curve which has four phases: the lag phase, the log or exponential growth phase, the stationary phase, and the death phase.

1. **Lag phase:** During this phase, there is no increase in cell number; rather, bacteria are preparing for reproduction and synthesizing DNA and various inducible enzymes needed for cell division.

2. **Log phase:** This phase, also called the exponential phase, follows the lag phase and starts with a rapid increase in bacterial number. At this stage, the logarithm of bacterial biomass increases linearly with time so that numbers of bacterial cells in a given interval of time is proportional to the biomass of bacteria present.

3. **Stationary phase:** The number of bacteria reaches a maximum in this phase and does not increase further (the growth rate is exactly equal to the death rate). This phase is sometimes called the plateau stage. A bacterial population may reach stationary growth

when a required nutrient is exhausted, when inhibitory end products accumulate, or when physical conditions are inappropriate for growth. The duration of the stationary phase varies, with some bacteria exhibiting very long stationary phases.

4. **Death phase:** Eventually, the number of viable bacterial cells begins to decline, signaling the onset of the death phase. No further divisions occur in this phase. Death rate, in many cases, follows the same kinetics as the exponential growth.

Principle:

Turbidity, which is a measure of the growth of microorganisms, is determined via an instrument called a spectrophotometer. The basis for the use of this instrument is that less light passes through a turbid solution than through a clear solution; i.e., the more turbid a solution (more growth), the less the amount of light transmitted through the solution. Since the optical density of the culture is proportional to the cell density, measuring the turbidity of the culture solution can be used in estimating numbers of bacterial cells, if a growth curve for the conditions used has already been established. This is the most common method used to rapidly estimate bacterial numbers.

Laboratory Supplies: 10 ml culture of an unknown microorganism, 50 ml LB broth in 125-ml flasks, Spectrophotometer, Un-inoculated LB broth in small test tube for blank.

Procedure:

1. Obtain a flask of LB and transfer 5 ml of broth to a small test tube. This is your blank. Adjust the wavelength to 550 nm on the spectrophotometer and calibrate the machine to 0.0 O.D. or 100% transmittance with your blank. Take the blank out but do not discard it.
2. Inoculate your flask with 10 ml of the culture provided and swirl the flask well.
3. Work quickly to withdraw 5 ml of the sample into a clean small test tube, using a pipette.

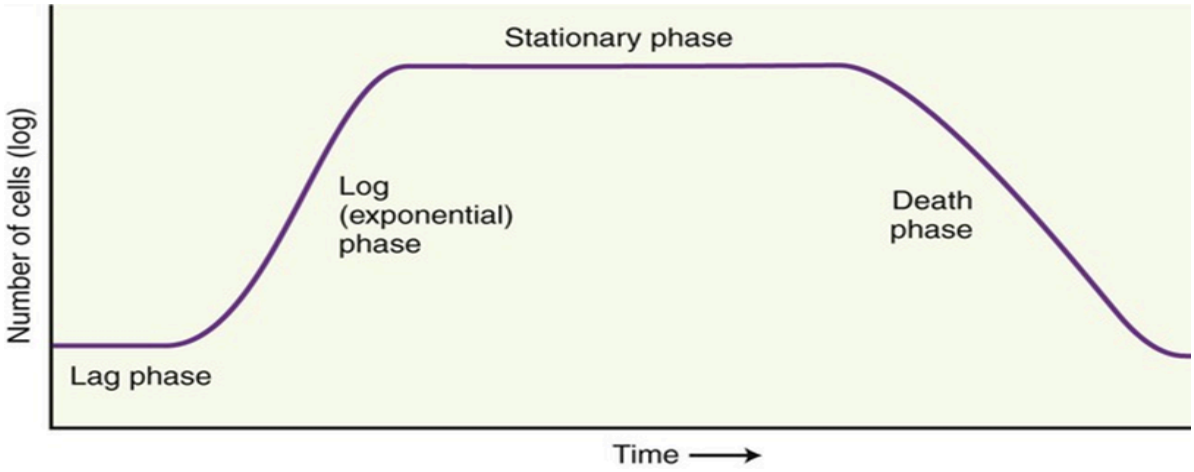
4. Place the culture flask immediately at the assigned incubating temperature and aeration condition and record the time.

5. Measure the O.D. of your sample next. Record the 0-time absorbance. Discard the contents of your sample test tube. This test tube can be re-used for your next sample without washing.

6. At 15-minute intervals for up to 90 minutes, shake your flask vigorously and transfer 5 ml of the culture to your sample test tube.

7. Record the optical density at each time period. Use your blank each time to make sure your readings are correct and other students have not altered the machine's calibration. Your instructor will ask all groups to write their results in a table on the blackboard. Copy these results to your Results Sheet. If more than one group is doing a growth condition, get their average data.

8. Plot the optical density (y-axis) versus time (x-axis) for each growth condition on the same regular graph paper.



Practical 9: Product recovery

Introduction

The recovery and purification of fermentation products is one of the most important aspects of industrial fermentation processes. The selection of suitable process of recovery and purification depends upon the nature of the end product, their concentration, the by-products present, the stability of the product and degree of purification.

However, the selected procedure must be quick, simple, reliable, accurate and must measure the desired compound accurately from among the different chemicals that may be present in the growth medium. It is generally considered that the fermentation and product recovery are integral parts of an overall process. Because of interaction between the two, neither of them be developed independently, is necessary as this must not result in problems and unnecessary expenditure.

General Pattern of a Recovery Process:

The following flow chart shows the various stages that are generally involved in the recovery process of a biological product. The recovery of an extracellular product of a fermentation broth is taken as an example to describe the various stages involved in a recovery process

The first stage for the recovery of an extracellular product is the removal of the large solid particles and microbial cells by centrifugation or filtration process. The broth is then fractionated or extracted into major fractions. Fractionation is carried out by employing either ultrafiltration or reverse osmosis or adsorption/ion exchange/gel filtration or affinity chromatography or liquid – liquid extraction or two-phase aqueous extraction based on chemical and physical properties of substances or precipitation.

The product-containing fraction is then purified by fractional precipitation. Further, more precise chromatographic techniques or crystallization process is employed to obtain a product, which is highly concentrated and devoid of any impurities

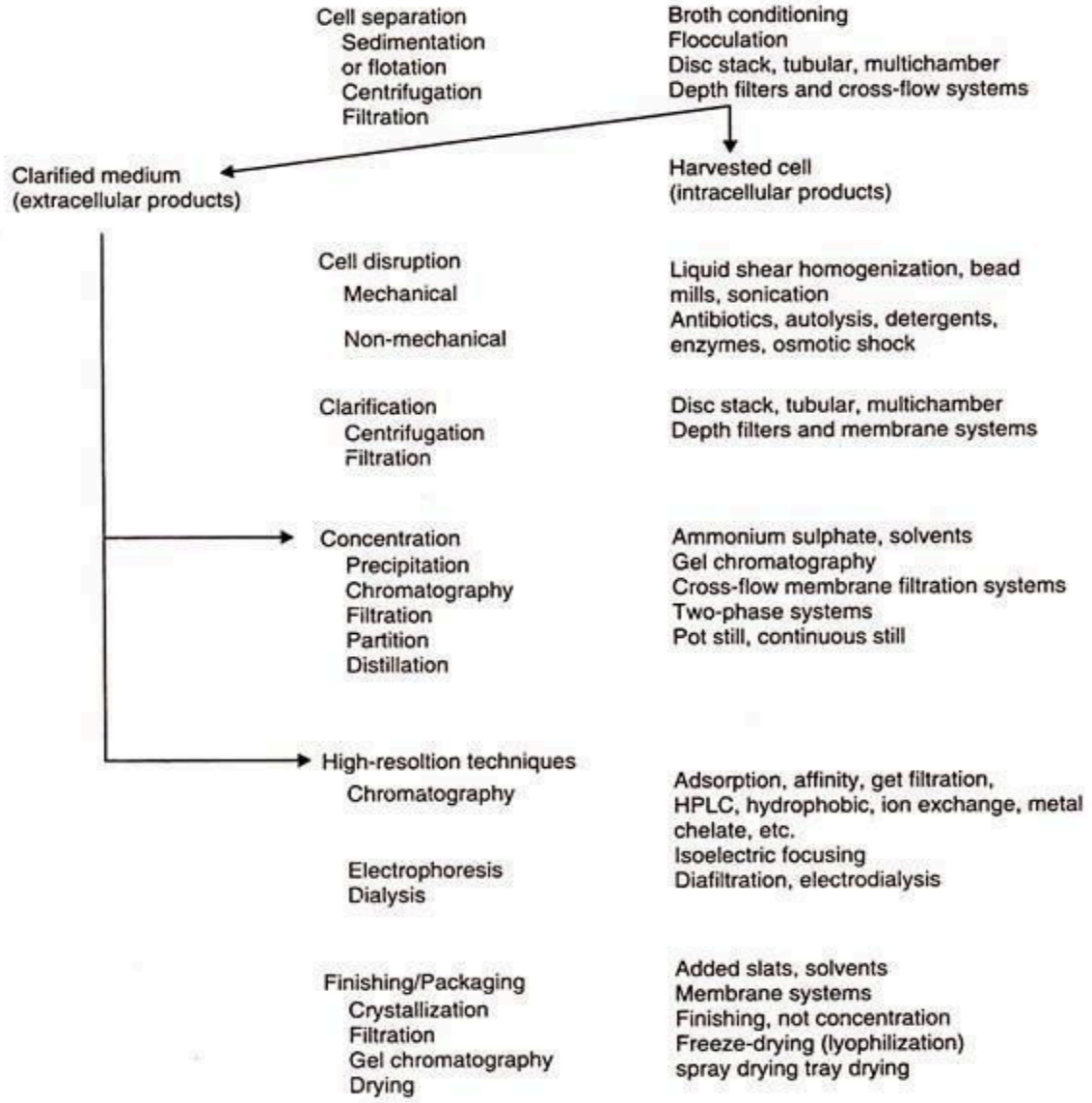


Fig. 2.23: Typical unit processes used in downstream processing.

Various recovery processes can broadly be classified into the following categories:

1. Physico-chemical processes.
2. Chromatographic partition processes.
3. Biological assays.

1. Physico-Chemical Processes:

Though there are many physico-chemical recovery processes employed for the isolation of fermentation products. However, it is to be remembered that the actual choice of recovery

process largely depends upon chemical reaction or chemical analysis involved because the fermentation broth contains many compounds in addition to the desired one.

(a) Turbidity Analysis:

(b) Titration and Gravimetric Analysis:

(c) Spectrophotometry Determinations:

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2. Chromatographic Partition Determinations:

The term chromatography was originally used in earlier times for a technique to separate coloured compounds. Although partition is the predominant cause of separation, other factors are also involved though in a minor fashion.

Employment of partition chromatography on paper or thin layer plates, not only made marked strides in the fermentation research and technology, but also allowed to detect and identify many types of new fermentation products even though they are in extremely smaller amounts. An unknown compound can easily be detected with the help of this technique by comparing the partition value with already known compounds.

The basic principle involved in the partition chromatography is the continuous partition of the solute or the sample between a stationary phase, such as paper or the silica gel thin layer plates, and a mobile phase consisting of a mixture of solvents, as these solvents move across the paper or silica gel layer.

Basically there are eight types of partition chromatography:

(i) Paper chromatography
(ii) Thin layer chromatography
(iii) Column Chromatography
(iv) Ion Exchange Chromatography
(v) Adsorption Chromatography
(vi) Affinity Chromatography
(vii) Gel Filtration
(viii) Gas Chromatography

3. Biological Assays:

Determining the effect of a fermentation product or an enzyme on the growth and metabolic activities of test organisms is called biological assay or bioassay. Therefore, it is an indirect method of assessing the quality of a fermentation product. Biological assays are, therefore, difficult to perform, less reproducible and usually do not give accurate results.

The microorganisms employed in these assays are called as test organisms. Strains of microorganisms occurring in nature or artificially mutated ones may be employed as test organism. Various categories of microorganisms are used as test organisms. For example, bacteria for assaying amino acids, antibiotics and vitamins; fungi for assaying vitamins and trace elements; yeasts for assaying vitamins and antibiotics.

A microorganism that is employed in a biological assay should possess the following characters:

1. It should genetically be stable and should not exhibit any genetic change in response to the test chemical.
2. It should respond in a graded manner to the test compound and not to other compounds.
3. It should not be a pathogen and should grow on simple medium.
4. It should preferably be aerobic or facultative aerobic but not an anaerobic.

5. It should grow well at a pH that does not affect the stability or toxicity of the fermentation product under assay.

Types of Biological Assays:

Biological assays can be categorized into four general groups.

(i) Diffusion assays,
(ii) Turbidimetric assays,
(iii) Metabolic response assays and
(iv) Enzymatic assays.